The Derivative from First Principles

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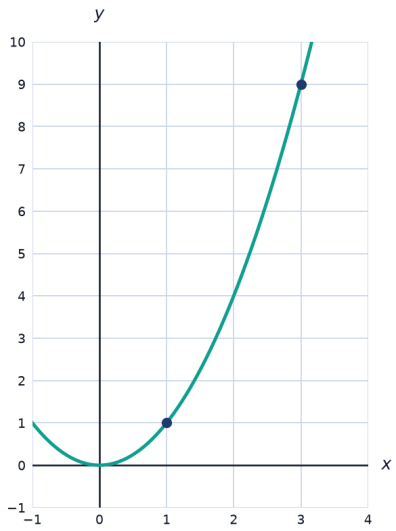
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The definition of the derivative**

Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.

2 **[Foundational]** • **Target: The definition of the derivative**

The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.

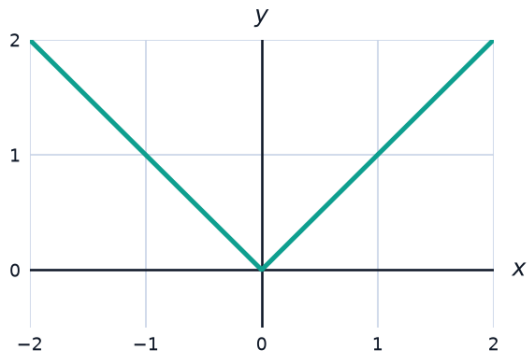


a Find the slope of the secant line.

b Find $f'(2)$ using the derivative, and compare it to the secant slope.

3 **[Foundational]** • **Target: Differentiability**

Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.



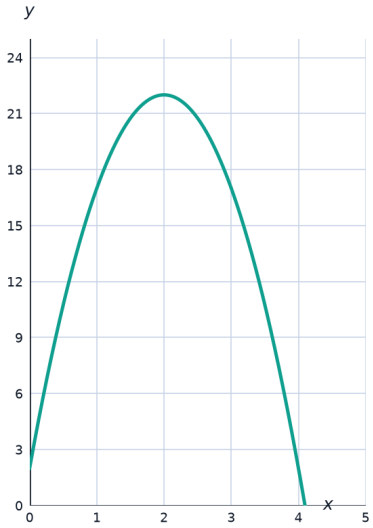
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4 [Foundational] • Target: Differentiability

State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?

5 [Foundational] • Target: Interpreting the derivative

A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.

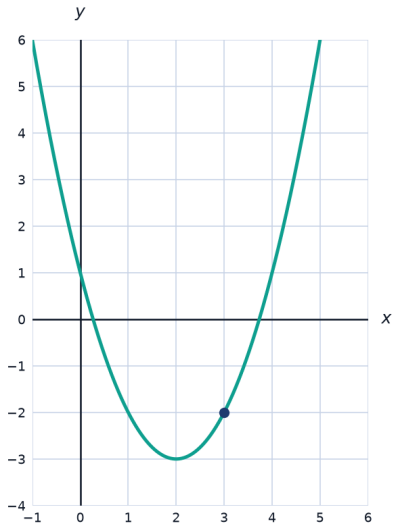


6 [Foundational] • Target: Point-derivatives from the definition

Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3 - 1$.

7 [Foundational] • Target: The derivative as the slope of the tangent line

For $f(x) = x^2 - 4x + 1$, use the derivative (found via the limit definition, $f'(x) = 2x - 4$) to find the equation of the tangent line at $x = 3$.

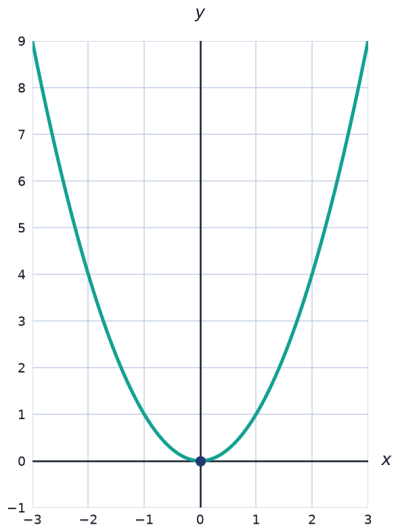


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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

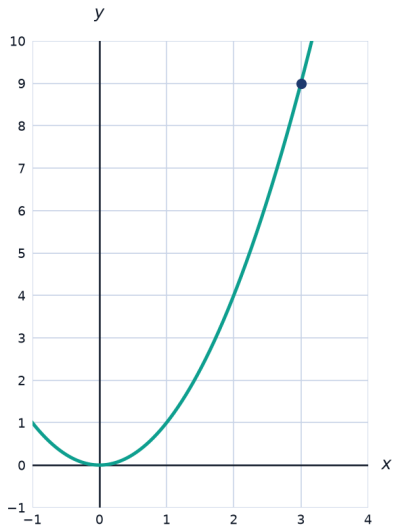
8 **[Intermediate]** • **Target: The derivative as the slope of the tangent line**

Explain why the tangent line to $y = x^2$ at $x = 0$ is horizontal, using the derivative $f'(x) = 2x$ found from the limit definition.



9 **[Intermediate]** • **Target: The derivative function versus the derivative at a point**

Explain the difference between $f'(x)$ and $f'(3)$, using $f(x) = x^2$ as an example.



10 **[Intermediate]** • **Target: Estimating a derivative from a table**

A table gives $f(1.9) = 8.61$, $f(2.0) = 9.00$, $f(2.1) = 9.41$. Use a symmetric difference quotient to estimate $f'(2)$.

11 **[Intermediate]** • **Target: The derivative of a constant and linear functions**

Use the limit definition to show that the derivative of a constant function $f(x) = c$ is 0.

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12 [Intermediate] • Target: The derivative of a constant and linear functions

Use the limit definition to show that the derivative of a linear function $f(x) = mx + b$ is the constant m.

13 [Intermediate] • Target: Interpreting the derivative

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

14 [Intermediate] • Target: Point-derivatives from the definition

Level 2 · MediumFind the point (x, y) on the parabola $y = x^2$ that is closest to the point $(0, 2)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 11.01 Sequences and Series

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ of $y = x^{\sin x}$ for $x > 0$ using logarithmic differentiation.

16 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 12.01 Convergence Tests

Level 1 · FoundationsFind the derivative of $y = \sqrt{x^4 + 3x^2 + 7}$ using the Chain Rule, and compute $y'(1)$.

17 [Challenging] • Target: Estimating a derivative from a table • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

18 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

19 [Challenging] • Target: The definition of the derivative • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

20 [Challenging] • Target: Differentiability • Cross-Lesson: 15.01 Double Integrals

Level 3 · AdvancedEvaluate the indeterminate limit using repeated applications of L'Hôpital's Rule with complete hypothesis verification: $\lim_{x \to 0} \frac{x - \arctan x}{x^3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The definition of the derivative

Tangent line slope to $y = \ln x$ at $x = e$:

a. 1

b. e

c. $1/e$

d. $\ln e$

22 [Foundational] • Target: Differentiability

The First Derivative Test classifies a critical point c as a local max when f' changes from:

a. positive to positive

b. negative to positive

c. zero to zero

d. positive to negative

23 [Foundational] • Target: More derivatives from the definition

A differentiable function $f(x)$ is such that its derivative $f'(x)$ is defined by $f'(x) = 3x^2 - 12x + 9$. The function $f(x)$ has a local maximum at $x = 1$ and a local minimum at $x = 3$. The graph of $y = f(x)$ passes through the point $(0, 5)$. Consider the tangent line to the graph of $y = f(x)$ at the point where $x = 0$. What is the y-intercept of this tangent line?

a. 5

b. 12

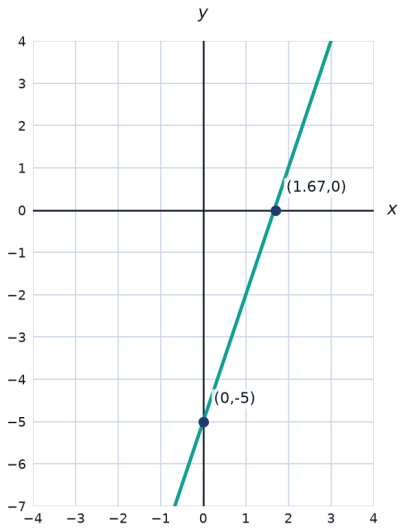
c. -3

d. 9

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24 [Foundational] • Target: Interpreting the derivative

The slope of $y = 3x - 5$ at any point is:



a. -5

b. 0

c. 3

d. undefined

25 [Foundational] • Target: Point-derivatives from the definition

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of t at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

a. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$

b. Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

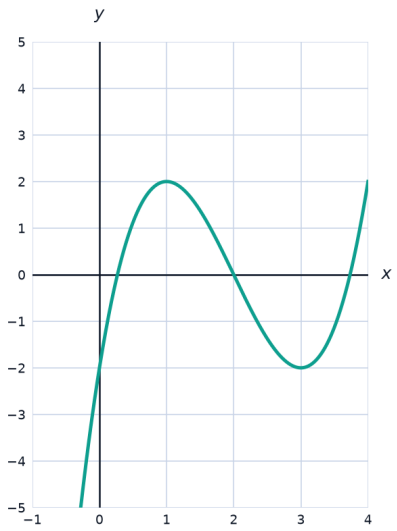
c. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

d. Rest at $t = \pi$, acceleration $a = 0$

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26 [Foundational] • Target: The derivative as the slope of the tangent line

Consider a function $f(x)$ whose derivative is given by $f'(x) = 6x^2 - 12x + 6$. The graph of the function $g(x) = x^3 - 6x^2 + 9x - 2$ is shown below. If $f(x)$ and $g(x)$ have the same tangent line at $x = 1$, determine the value of $f''(0)$.



a. -6

b. -12

c. 0

d. 6

27 [Foundational] • Target: The derivative function versus the derivative at a point

Find the slope of the tangent line to $r = 2\cos(\theta) + 2$ at $\theta = \frac{\pi}{4}$.

a. $-\sqrt{2}$

b. $1 - \sqrt{2}$

c. $1 + \sqrt{2}$

d. 1

28 [Foundational] • Target: Estimating a derivative from a table

Parametric tangent at $t = t_0$: slope $m = \frac{y'(t_0)}{x'(t_0)}$. If $x'(t_0) = y'(t_0) = 0$, the point is:

a. always a cusp

b. regular

c. a node

d. a singular/cusp point — need L'Hôpital

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: The derivative of a constant and linear functions**

Prove function is constant: if $f'(x) = 0$ on (a, b), then by MVT:

a. $f(x) = f(a)$ for all x

b. f is zero

c. f is periodic

d. f is odd

30 **[Intermediate]** • **Target: The definition of the derivative**

Local min at $x = c$ by First Derivative Test requires f':

a. positive then negative

b. negative then positive

c. positive everywhere

d. zero everywhere

31 **[Intermediate]** • **Target: Differentiability**

At which point does $f(x) = x^2 - 4x$ have a horizontal tangent?

a. $x = 0$

b. $x = 2$

c. $x = 4$

d. $x = -2$

32 **[Intermediate]** • **Target: More derivatives from the definition**

The slope of the tangent to $f(x) = x^2$ at $x = 3$ is:

a. 3

b. 6

c. 9

d. 12

33 **[Intermediate]** • **Target: Interpreting the derivative**

The derivative of $f(x) = \frac{x^4 - 1}{x^2 + 1}$ at $x = 1$ using limit definition:

a. 1

b. 2

c. 3

d. 4

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34 [Intermediate] • Target: Point-derivatives from the definition

The Second Derivative Test says $f''(c) < 0$ at a critical point c implies:

a. inflection point

b. no conclusion

c. local maximum

d. local minimum

35 [Intermediate] • Target: The derivative as the slope of the tangent line

Acceleration is the second derivative of position. If $a(t) = g$ (constant), integrating twice gives:

a. $s(t) = \frac{1}{2}gt^2 + v_0t + s_0$

b. $s(t) = gt$

c. $s(t) = v_0t$

d. $s(t) = g + v_0$

36 [Intermediate] • Target: The derivative function versus the derivative at a point

The tangent line to $y = \ln x$ at $x = e$ has equation:

a. $y = x/e$

b. $y = x/e - 1 + 1$

c. $y = \frac{1}{e}(x - e) + 1$

d. $y = e(x - e) + 1$

37 [Intermediate] • Target: Estimating a derivative from a table

The normal line to $y = x^2$ at $x = 1$ has slope:

a. -2

b. $-\frac{1}{2}$

c. $\frac{1}{2}$

d. 2

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The derivative of a constant and linear functions • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The second derivative test: $D = f_{xx}f_{yy} - f_{xy}^2$. If $D > 0$ and $f_{xx} > 0$, the point is a:

a. saddle point

b. local min

c. local max

d. inflection point

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39 [Challenging] • Target: The definition of the derivative • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Jerk = third derivative of position. If jerk is constant j, acceleration is:

a. $jt + a_0$

b. jt^2

c. j/t

d. e^{jt}

40 [Challenging] • Target: Differentiability • Cross-Lesson: 15.01 Double Integrals

The graph of a function $g(x)$ is shown above. The function $g(x)$ is defined by $g(x) = \frac{1}{3}x^3 - ax^2 + 2x + 1$, where a is a positive constant. The graph has a tangent line at $x = 1$ which is parallel to the line $y = -2x + 5$. Find the value of the constant a.

a. 1

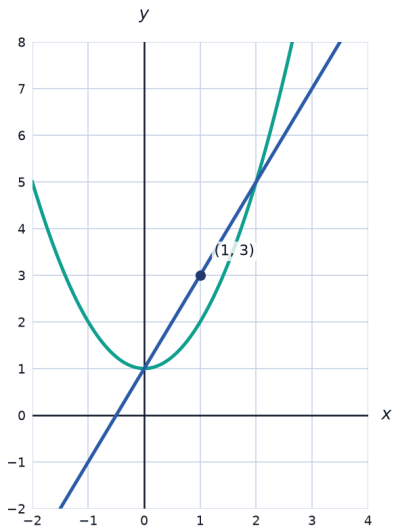
b. 2

c. 3

d. 4

41 [Challenging] • Target: More derivatives from the definition • Cross-Lesson: 16.01 Vector Fields and Line Integrals

The graph of a differentiable function $f(x)$ is shown below. The tangent line to the graph at $x = 1$ has the equation $y = 2x + k$. Given that $f(1) = 3$, determine the value of k.



a. 1

b. 2

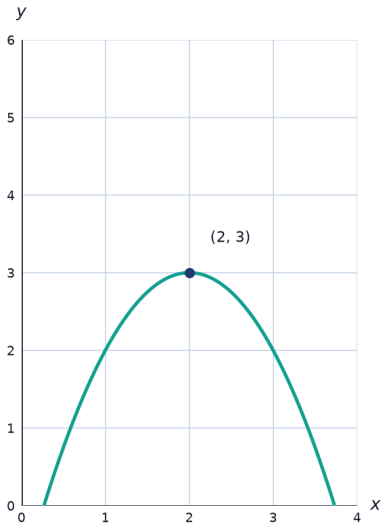
c. 3

d. 4

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42 [Challenging] • Target: Interpreting the derivative • Cross-Lesson: 2.01 Evaluating Limits

The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.



a. 4

b. 5

c. 6

d. 7

43 [Challenging] • Target: Point-derivatives from the definition • Cross-Lesson: 2.02 Continuity

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

44 [Challenging] • Target: The derivative as the slope of the tangent line • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Tangent line to $r = \theta$ at $\theta = \pi$:

a. $y = -x$

b. horizontal

c. slope π

d. slope $-\pi$

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45 [Challenging] • Target: The derivative function versus the derivative at a point • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Unit tangent vector $\mathbf{T}(t) = \mathbf{r}'(t)/|\mathbf{r}'(t)|$. It has magnitude:

a. 1

b. 0

c. $|r'|$

d. t

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